National Institute of Technology Karnataka, Surathkal

Ambulance Booking System

User Manual

For patients, ambulance crews and the control room

This manual explains what each of the three roles can do, and how. It follows the order somebody meets the app in: signing in first, then the task that brought them to it.

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1 Before you start

One app serves all three roles. Which one you get is decided by the account you sign in with, not by a separate download.

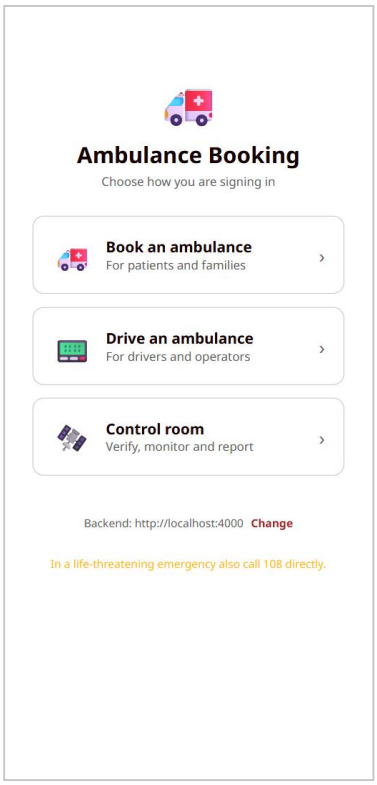


Figure 1 — the first screen.

What you need

- An Android phone, and the **Ambulance Booking** app installed on it.
- An email address you can open on that phone — the app sends a code to it when you register, and you cannot finish without reading that code.
- A mobile number. It is not used to sign in; it is what the ambulance crew ring if they cannot find you.
- A working internet connection. The app talks to a server for everything.

Choose the role you are signing in as. If you choose the wrong one, go back and choose again — nothing is decided until you sign in.

The three roles

Role	What it is for
Book an ambulance	Patients and their families. Request an ambulance, watch it approach, pay, and rate the trip.
Drive an ambulance	Drivers and operators. Go on duty, accept requests, move a trip through its stages, see earnings.
Control room	Administrators. Verify ambulances, watch every live booking, read reports and run the figures.

> **In a real emergency**
> This app is a demonstration built for a university course. In a life-threatening emergency, call **108** directly. The app says the same thing on every screen it can.

2 Getting an account

Registering proves your email address. Nothing is created until you answer the code that is sent to it, so an account always has an address behind it that somebody has actually read.

2.1 Registering

1. Open the app and choose your role.
2. Tap **Register**.
3. Enter your full name, mobile number, email address and a password.
4. Tap **Email me a code**. Nothing has been created yet at this point.
5. Open your email and find the six-digit code. It is valid for five minutes.
6. Type the code into the app and tap **Verify and create account**.

You are signed in as soon as the code is accepted. If you register as a driver or an operator, there is one more step — see section 4.1.

Choosing a password

The app refuses a password that is easy to guess, and tells you why as you type rather than after you submit. It will not accept one that is shorter than eight characters, appears on the list of most commonly used passwords, is one character repeated or a simple run, or contains your own name, number or email address.

Either a long passphrase or a shorter mix of letters, digits and symbols will do. "correct horse battery staple" is accepted; "Password1!" is not.

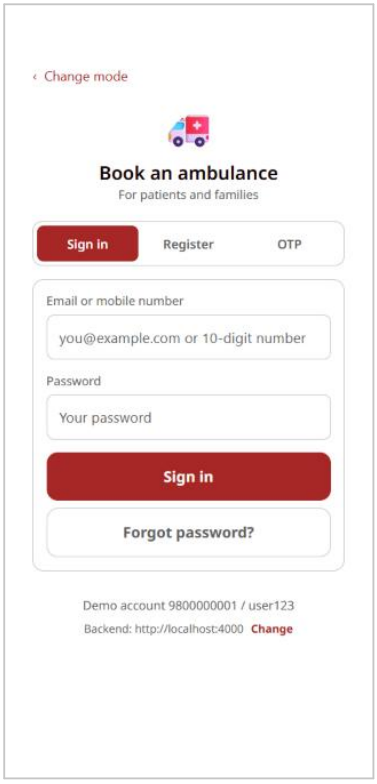


Figure 2 — signing in.

2.2 Signing in

- **With a password** — enter your email address or your mobile number, and your password.
- **With a code** — tap **OTP**, enter your email address, and the app emails you a code to sign in with. Useful if you have forgotten your password and do not want to reset it.

2.3 If you forget your password

1. On the sign-in screen, tap **Forgot password?**
2. Enter your email address and tap **Email me a reset code**.
3. Enter the code from your email and choose a new password.

> **Resetting signs you out everywhere**
> Every session on every device ends. That is deliberate: if somebody else knew your password, leaving them signed in would defeat the reset.

3 For patients — booking an ambulance

Four tabs: **Book**, **Track**, **Trips** and **Account**.

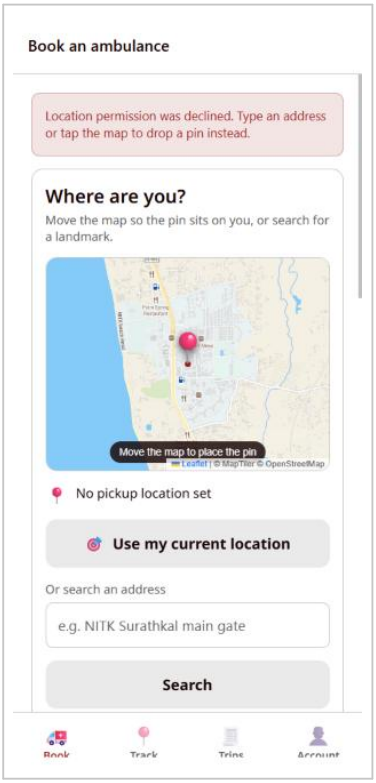


Figure 3 — the booking screen.

3.1 Setting where you are

This is the most important thing on the screen, so the app offers three ways to do it and you can use whichever is quickest.

- **Use my current location** — reads your GPS. The quickest, when it works. If a fix is slow the app uses your last known position rather than waiting for ever.
- **Move the map** — the pin stays in the middle of the map while the map moves under it. Put the pin where you are and the address below updates.
- **Search an address** — type a landmark, such as NITK Surathkal main gate, and pick from the results.

Get this right before anything else

The address under the pin is where the ambulance is sent. Check it reads somewhere you recognise before you request.

3.2 Choosing the kind of ambulance

Choose	When
EMG — Emergency	You need an ambulance now and do not know which kind. Every ambulance nearby is alerted at once and the first to accept comes.
BLS — Basic Life Support	Oxygen, a stretcher and a trained attendant.
ALS — Advanced Life Support	Ventilator, defibrillator and a paramedic.
ICU — Mobile ICU	Critical care with full monitoring.
PTV — Patient Transport	A planned, non-emergency transfer.

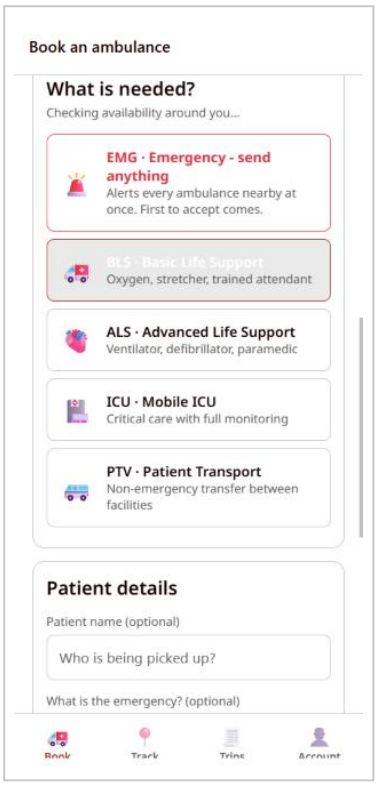


Figure 4 — the five kinds.

If you are unsure, choose Emergency. The other four go to one ambulance at a time, in order; an emergency goes to all of them together, which is faster.

3.3 Requesting it

1. Add the patient name and a short note if you can — the note reaches the crew with the request, and "chest pain, conscious" tells them more than nothing does.
2. Check the indicative fare. The final invoice is worked out from the distance actually travelled, so it will not match exactly.
3. Tap **Request**.

The app moves to **Track** while it looks for an ambulance. Each one has 45 seconds to answer before the request moves to the next.

3.4 Watching it come

Once an ambulance accepts, **Track** shows it moving on the map with an estimated arrival time that updates as it travels, the driver's name and vehicle, and the trip stages as they are reached.

- **Call driver** puts you through on the phone. Use it if they cannot find you.
- **Cancel** is available until the patient is on board.

3.5 Afterwards

- The invoice appears under **Trips**, itemised: base fare, distance, any surcharge and tax.
- Pay by UPI, card, net banking or cash on completion.
- Rate the trip from one to five stars, with a comment if you want.
- If something was wrong, you can report it — see section 6.

4 For ambulance crews — the dispatch console

Four tabs: **Console**, **Trip**, **Earnings** and **Profile**.

4.1 Adding your ambulance

A new driver or operator cannot receive anything until the control room has seen their vehicle. Straight after registering you are asked for it.

1. Enter the registration number as printed on the plate, without spaces.
2. Choose what the ambulance is: BLS, ALS, ICU or PTV.
3. Add the licence number and a link to scanned documents if you have them.
4. Tap **Submit for verification**.

> **Until an administrator verifies you**
>
> The console stays locked and you will receive no requests at all. It unlocks by itself the moment the control room verifies you — you do not need to sign out and back in, and you do not need to keep checking.

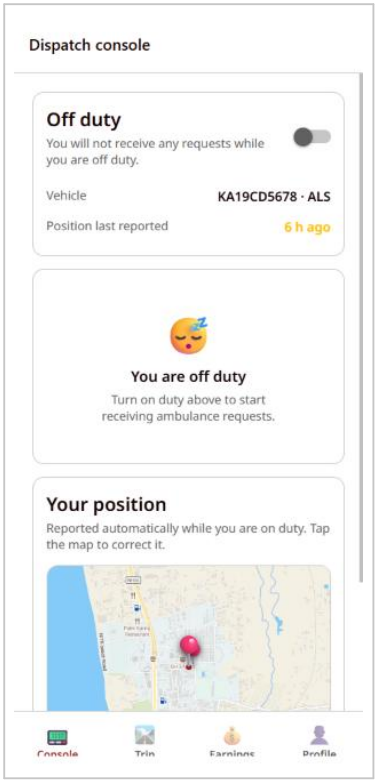


Figure 5 — the console, off duty.

4.2 Going on duty

Use the switch at the top of **Console**. While you are on duty the app reports your position about every twenty seconds, which is how you are matched to somebody nearby.

An ambulance whose app has stopped reporting for fifteen minutes is taken off duty automatically, because the system cannot send a patient to a vehicle it cannot locate. If that happens you are told, and you simply go on duty again.

4.3 Answering a request

A request arrives as a card with a countdown, and your phone sounds and vibrates. You have **45 seconds**; after that it goes to the next ambulance.

- A card marked **EMERGENCY** went to every ambulance nearby at once. The first to accept takes it, so hesitating costs you the job rather than delaying it.
- Declining passes it straight on. It does not count against you.

> **Alerts only work while the app is running**
>
> Backgrounded is fine — your phone will still sound. But if the app is swiped away from recent apps, nothing can reach it. Leave it running while you are on duty.

4.4 Running the trip

The **Trip** tab shows the pickup, the route and your own position, and moves through the stages one at a time. You cannot skip one.

Stage	Tap when
On the way	You set off towards the pickup.
Arrived	You reach the pickup point.
On board	The patient is in the ambulance. **The fare starts counting from here.**
Complete	The patient has been handed over. The invoice is raised.

> **On board and Complete decide the fare**
> The distance billed is measured between those two taps, from your GPS — not from anything the app is told. Tapping **On board** late, or **Complete** early, costs you the difference.

4.5 Earnings

The **Earnings** tab totals what has settled and what is outstanding, with a fortnight's history and every past trip. Each trip there can also be reported — see section 6.

5 For the control room

Five tabs: **Overview**, **Bookings**, **Fleet**, **Providers** and **More**. Everything updates live.

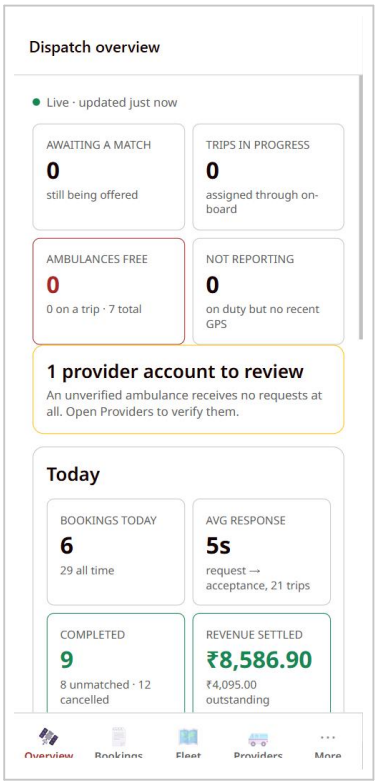


Figure 6 — the dispatch overview.

5.1 Overview

The operational picture, refreshing itself as things happen. Nothing here needs to be reloaded.

- **Awaiting a match** — requests still being offered to ambulances.
- **Trips in progress** — assigned, through to on board.
- **Ambulances free** — verified, on duty, and reporting position: the ones that could actually be dispatched right now.
- **Not reporting** — on duty but their app has gone quiet. These need chasing: they cannot be dispatched.

5.2 Verifying an ambulance

1. Open **Providers** and filter by **PENDING**.
2. Open the one you are checking and read the registration, licence and documents.
3. Choose **Verify** or **Reject**, with a remark saying why.

The driver is told at once and their console unlocks on its own. An unverified ambulance receives no requests at all, so this is what puts a vehicle into service.

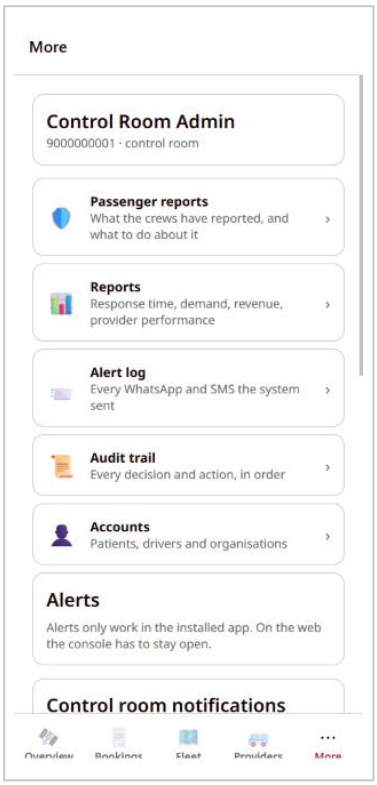


Figure 7 — the control room menu.

5.3 Watching and rescuing bookings

- **Bookings** lists everything, filterable by status, type or date.
- Opening one shows the full case file: who asked, every ambulance offered it and what they said, the GPS trace, the invoice, the rating and every alert sent.
- **Re-dispatch** runs matching again on a booking that failed.
- **Force cancel** closes one that is stuck.

5.4 Everything else

The rest of the control room sits behind **More**, listed below.

Under More	What it holds
Passenger reports	Complaints from crews and from patients, and what to do about them. See section 6.
Reports	Response time, demand by type, revenue and provider performance over a date range.
Alert log	Every WhatsApp, email and push the system has sent. Fills live.
Audit trail	Every decision and action in order, including your own.
Accounts	Every patient, driver and operator. Suspend or reinstate.

6 Reporting somebody

A hoax call takes an ambulance somebody else needed, and a crew who behaves badly should answer for it. Either side can report the other, and the process is the same both ways.

Reporting is not a punishment

Filing a report changes nothing about the other person's account. It goes to the control room, who decide. Nobody is suspended because you say so.

6.1 A crew reporting a passenger

1. Open **Earnings** and find the trip.
2. Tap **Report this passenger**.
3. Choose what happened: hoax call, did not turn up, abusive, refused to pay, or something else.
4. Add what happened — brief and factual is best — and send it.

6.2 A patient reporting a crew

1. Open **Trips** and find the trip.
2. Tap **Report this ambulance**.
3. Choose what happened: never arrived, unsafe driving, rude or abusive, asked for more than the fare, or the state of the vehicle.
4. Add what happened and send it.

In both directions you can only report a trip you were part of, only once, and only once it is over.

6.3 How the control room rules on it

Open **Passenger reports** under **More**. Each one shows who it is about, who said so, what they said, and — the part that matters — **how many other reports name the same person, and how many were upheld**. One complaint is an incident; four are a pattern.

Decision	What it does
Dismiss	No action. A report found baseless costs the person nothing, so no penalty can be attached to a dismissal.
Uphold — warn	Recorded against them, nothing more.
Uphold — suspend	Signed out everywhere and locked out. A suspended driver is also taken off duty, so no further requests reach them. Reversible from **Accounts**.
Uphold — delete	Name, number and email erased and the account locked. Their bookings, invoices and the report itself are kept — the record has to outlive the person. Not reversible.

Whoever filed the report is told the outcome either way.

7 If something goes wrong

What you see	What to do
The server did not answer within a minute	The backend sleeps when unused and takes about fifty seconds to wake. Try again — it will be quick the second time.
Cannot reach the server	Check the phone is online. If you are running your own backend, check the address on the sign-in screen under **Backend … Change**.
No GPS fix yet	Move somewhere with a clearer view of the sky, or set the position on the map instead. The app will not wait for ever.
Your code never arrives	Check the spelling of the address and look in spam. A made-up domain will never deliver.
No ambulance could be assigned	None was free within fifteen kilometres. Try again, choose **Emergency** so every nearby ambulance is asked at once, or call 108.
The driver console stays locked	The ambulance has not been verified yet. It unlocks on its own; nothing needs restarting.
No alert when a request arrives	Open **Profile**, then **Alerts**, and tap **Send a test alert**. Alerts need the app to be running — backgrounded is fine, closed is not.
Your account is suspended	An administrator has suspended it after a report. Contact the control room.

Demonstration accounts

The seeded database ships with these, for trying the app without registering.

Role	Sign in with
Patient	9800000001 / user123
Driver (ALS)	9700000002 / driver123
Driver (ICU)	9700000003 / driver123
Control room	9000000001 / admin123

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